Bias and Fairness Report

- Possible gaps in data – the training data comes from just one highway, and not the entire city street network. Because of this, the model understands the highway commuting weel, but that may not be suitable for city streets, intersections, neighbourhood roads etc. The rare weather events like major storms or holiday accidents also appear very less in the data, giving the model fewer data to learn them.
- The model estimates the traffic danger using weather and car counts instead of direct accident reports. If a blocked road due to road works or a car which has broken down blocks a lane on a clear day, the model can easily miss the slowdown caused.

Governance and Sustainability in Mobility AI

- Safety Checks Before Real-World Use:
  - A human review has to be introduced to review and make traffic adjustments, instead of automating the flow immediately. An elaborate test should be conducted before automation.
  - Data monitoring has to be real-time, there has to be automated checks to constantly watch the weather and traffic numbers, so that the checks catches bad data before it affects recommendations.
  - Every deployed model must be saved and tracked in a central registry like MLFlow, so the teams can find which exact version made a certain prediction.
- Resource and Environmental Impacts:
  - Encouraging and advising commuters to travel outside the busy hours cuts road blocks and congestion, and hence waiting, helping lower vehicle fuel consumption and pollution.